

Conclusion

Summary of Contributions

This paper provided comprehensive simulation-based empirical validation of the Cognitive Integrity Framework (CIF) introduced in Part 1 of this series. Our contributions span implementation, evaluation, and analysis:

Implementation: We implemented the complete CIF defense suite—cognitive firewalls, belief sandboxes, trust calculus with bounded delegation, tripwire detection, behavioral invariants, and Byzantine-tolerant consensus—as production-ready Python modules with 1,557 passing tests at 100% pass rate, demonstrating that the formal mechanisms translate into deployable, independently testable code.¹

Attack Corpus: We assembled 950 cognitive attacks across four categories (prompt injection, trust exploitation, belief manipulation, coordination attacks), enabling reproducible security evaluation of multiagent systems.

Cross-Architecture Evaluation: We evaluated CIF's detection architecture across six production multiagent topologies (Claude

Key Findings

The simulation-based evaluation yields four principal findings, each reflecting CIF's design-level detection properties under calibrated conditions:

1. **Layered defense is essential:** No single mechanism achieves acceptable protection in simulation; composition yields multiplicative improvement consistent with theoretical predictions from the defense composition algebra.
2. **Trust calculus prevents amplification:** The δ^d decay bound successfully prevented trust laundering across all tested architectures—a structural guarantee that holds independent of attacker sophistication and is verified both formally (Part 1) and through unit-tested implementation.
3. **Architecture matters:** Peer-to-peer architectures show greatest improvement from CIF in simulation, consistent with Part 1's prediction that equal-trust topologies are most vulnerable to lateral movement attacks.

Observed Deployment Properties

The evaluation data establishes four empirical properties relevant to deployment:

1. **Layered defense is necessary for high efficacy:** No single mechanism exceeded 85% detection. The Minimal-C configuration (Firewall + Tripwires + Drift) achieved 90% at 12% overhead; full CIF reached 94% at 20–25% overhead (tab:minimal-configs).
2. **Defense efficacy is architecture-dependent:** Tripwire-only deployments achieved 82% detection in hierarchical topologies but only 61% in peer-to-peer systems. Trust calculus with $\delta \leq 0.8$ was the dominant factor in peer-to-peer defense (tab:architecture-insights).
3. **Detection degrades against novel attacks:** Cross-validation with held-out attack types showed 4–10% detection rate gaps, with coordination attacks exhibiting the largest generalization gap (–10%) (sec:robustness).

Alignment with Emerging Standards

CIF's design anticipates and directly addresses the security risks codified by two major 2025–2026 standardization efforts.

The **OWASP Top 10 for Agentic Applications** (2026) identifies 10 agentic-specific risks (ASI01–ASI10) [?]. CIF's defense mechanisms map systematically to these risks: the Cognitive Firewall counters Agent Goal Hijack (ASI01) by detecting and filtering prompt injections before they alter agent objectives; the Belief Sandbox addresses Tool Misuse and Exploitation (ASI02) by isolating unverified tool outputs before they propagate into the agent's belief state; Trust Calculus with δ^d decay prevents Identity and Privilege Abuse (ASI03) by enforcing bounded delegation depth and decaying trust across privilege boundaries; Tripwire monitoring detects Memory and Context Poisoning (ASI06) by alerting on unauthorized belief modifications; and Byzantine Consensus mitigates Cascading Failures (ASI08) by requiring supermajority agreement before collective actions, preventing a single compromised agent from triggering system-wide degradation. This mapping demonstrates that CIF provides a unified formal

Paper Series

This is Part 2 of the *Cognitive Security for Multiagent Operators* series:

- ▶ **Part 1: Formal Foundations** - Trust calculus, defense composition algebra, information-theoretic bounds
- ▶ **Part 2 (This Paper): Computational Validation** - Implementation, attack corpus, empirical results
- ▶ **Part 3: Practical Guidance** - Deployment checklists, operator posture, risk assessment

Together, these papers provide a complete framework for understanding, implementing, and operating cognitive security in multiagent AI systems.

Data and Code Availability

The CIF implementation (defense mechanisms, evaluation framework, analysis scripts) is available at [<https://github.com/docxology/cognitive_integrity>](https://github.com/docxology/cognitive_integrity) (DOI: 10.5281/zenodo.18364128). A sanitized subset of the attack corpus suitable for reproducibility is included; the full corpus is available to verified researchers upon request (see [sec:access-request](#)). All figures, tables, and statistical analyses can be reproduced using the provided scripts with the fixed random seed (42).

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Author Contributions

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Competing Interests

The authors declare no competing interests.

Ethics Statement

This research was reviewed and determined exempt from IRB oversight as it did not involve human subjects. All attacks were tested against synthetic agent configurations in sandboxed environments. Novel attack vectors were disclosed to affected framework maintainers following a 90-day responsible disclosure policy. Dual-use risks are mitigated through sanitization of published examples and restricted access to the full attack corpus (see sec:dual-use).